Chapter 2 - Upper body activity classification using an inertial measurement unit in court and field-based sports: a systematic review.

Preface

A systematic literature review was conducted to explore past research that classified upper body movements in court and field-based sports using inertial measurement units (IMU). This aided future chapters of the thesis by identifying equipment and techniques that worked successfully and possible gaps in the literature. It also provided the sports science field with a review that explained in a non-technical way how these systems work, as well as the benefits and limitations that need to be considered when using IMUs.

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Abstract

Inertial measurement units (IMUs) are becoming increasingly popular in activity classification and workload measurement in sports. This systematic literature review focuses on upper body activity classification in court or field-based sports. The aim is to provide sports scientists and coaches with an overview of the past research in this area and the processes and challenges involved in activity classification. The SPORTDiscus, PubMed, and Scopus databases were searched, resulting in 20 articles. Both manually defined algorithms and machine learning approaches have been used to classify IMU data with varying degrees of success. Manually defined algorithms may offer simplicity and reduced computational demand, whereas machine learning may benefit complex classification problems. Inter-study results show that no one machine learning model is best for activity classification; differences in sensor placement, IMU specification, and pre-processing decisions can affect model performance. Accurate classification of sporting activities could benefit players, coaches, and team medical personnel by providing an objective workload estimate. This could help prevent injuries, enhance performance, and provide valuable data to the coaching staff.

Key points

- IMUs, combined with manually-defined algorithms or machine learning, can classify upper body movements in court or field-based sports with varying levels of success.
- The number of IMUs, their placement location, and the sensor specifications need to be considered when developing a classification system, as these can affect accuracy.
- Classification accuracy is also affected by data processing (e.g., filtering, windowing, feature engineering) and the classification model used, regardless of the sport or action performed.
- As movement patterns can vary between elite and sub-elite athletes, a broad range of skill levels should be considered when training an algorithm for improved generalisation.

Introduction

Activity classification is a fundamental part of movement analysis for most athletes and coaches. It is important for enhancing performance, athlete monitoring, and injury prevention. Movement data for activity classification is often captured using video analysis, which can be expensive, have low portability, and be labour intensive to analyse.[46,47] In addition, the placement or angle of video cameras can lead to missed body movements or orientation issues.[48-50] An alternative approach that may overcome some of these limitations is using an inertial measurement unit (IMU). IMUs are wearable sensors that contain an accelerometer, gyroscope, and sometimes a magnetometer. The accelerometer measures linear acceleration (measured in g-force), the gyroscope measures angular velocity (degrees per second), and the magnetometer measures the strength and direction of the local magnetic field. This information can be used for activity classification, analysis of movement parameters, and measurement of physical load.

The IMU (including battery, processing unit, wireless unit and local storage) can now be made smaller than a matchbox, allowing comfortable attachment with minimal disturbance to normal movement.[51] The price can vary; however, consumer-grade IMUs can cost less than 100 USD, making them affordable for most athletes in developed nations. IMUs embedded in smartphones have recently been used for activity classification in basketball,[52] cross country skiing,[53] and soccer and hockey.[51] As smartphone subscriptions are expected to rise to 8.3 billion in 2023,[54] IMUs could be accessible to most of the world's population.

The present review focuses on research that classifies upper body movements in court or field-based sports using an IMU. The reason for this scope is that upper body movement patterns can have high variability between sports (e.g., cricket batting strokes, volleyball spike), therefore, providing additional challenges for classification. Lastly, this review will provide sports scientists and strength and conditioning coaches with an overview of the processes involved in activity classification and the various study design and data processing decisions that can affect classification performance. This information may assist with judging the applicability of IMU-based activity classification in different sporting contexts.

Methods

Search strategy

This systematic literature review followed the PRISMA guidelines. The databases PubMed, SPORTDiscus, and Scopus, were searched for articles dated from January 1st, 2000, to December 20th,

2018. The terms used for each database search can be observed in Table 2-1. The inclusion criteria consisted of full-length journal articles or published conference papers written in English. Articles must have presented classification results for an upper body movement in court or field-based sports using an IMU.

Table 2-1: Database search terms used.

PubMed
(((((((wearable OR microsensor OR device OR IMU OR inertial measurement unit OR triaxial OR sensor OR accelerometer OR gyroscope)) AND ((Movement detection OR event classification OR classification OR workload OR recognition OR analysis)) AND (cricket OR baseball OR softball OR handball OR American football OR discus OR volleyball OR tennis OR table tennis OR badminton OR throwing)) NOT (clinical OR review))))
SPORTDiscus
(wearable OR microsensor OR device OR IMU OR inertial measurement unit OR triaxial OR sensor OR accelerometer OR gyroscope) AND (Movement detection OR event classification OR classification OR workload OR recognition OR analysis) AND (cricket OR baseball OR softball OR handball OR American football OR discus OR volleyball OR tennis OR table tennis OR badminton OR throwing) NOT (clinical OR review)
Scopus
wearable OR microsensor OR device OR IMU OR "inertial measurement unit" OR triaxial OR sensor OR accelerometer OR gyroscope AND "Movement detection" OR "event classification" OR classification OR workload OR recognition OR analysis AND cricket OR baseball OR softball OR handball OR "American football" OR discus OR volleyball OR tennis OR table AND tennis OR badminton OR throwing AND NOT clinical OR review

Results

Selection process

A flow diagram of the process used to identify and select articles is detailed in Figure 2-1. The initial search resulted in 1,454 articles. The titles of these articles were then screened for duplicates and relevance to the inclusion criteria. In total, 79 article abstracts were screened, of which 43 were reviewed in their entirety. Out of these articles, 13 met the inclusion criteria. The reference lists of these articles were then reviewed, which netted an additional seven articles. This brought the total number of included articles to 20.

Figure 2-1: PRISMA flow diagram of the systematic search.

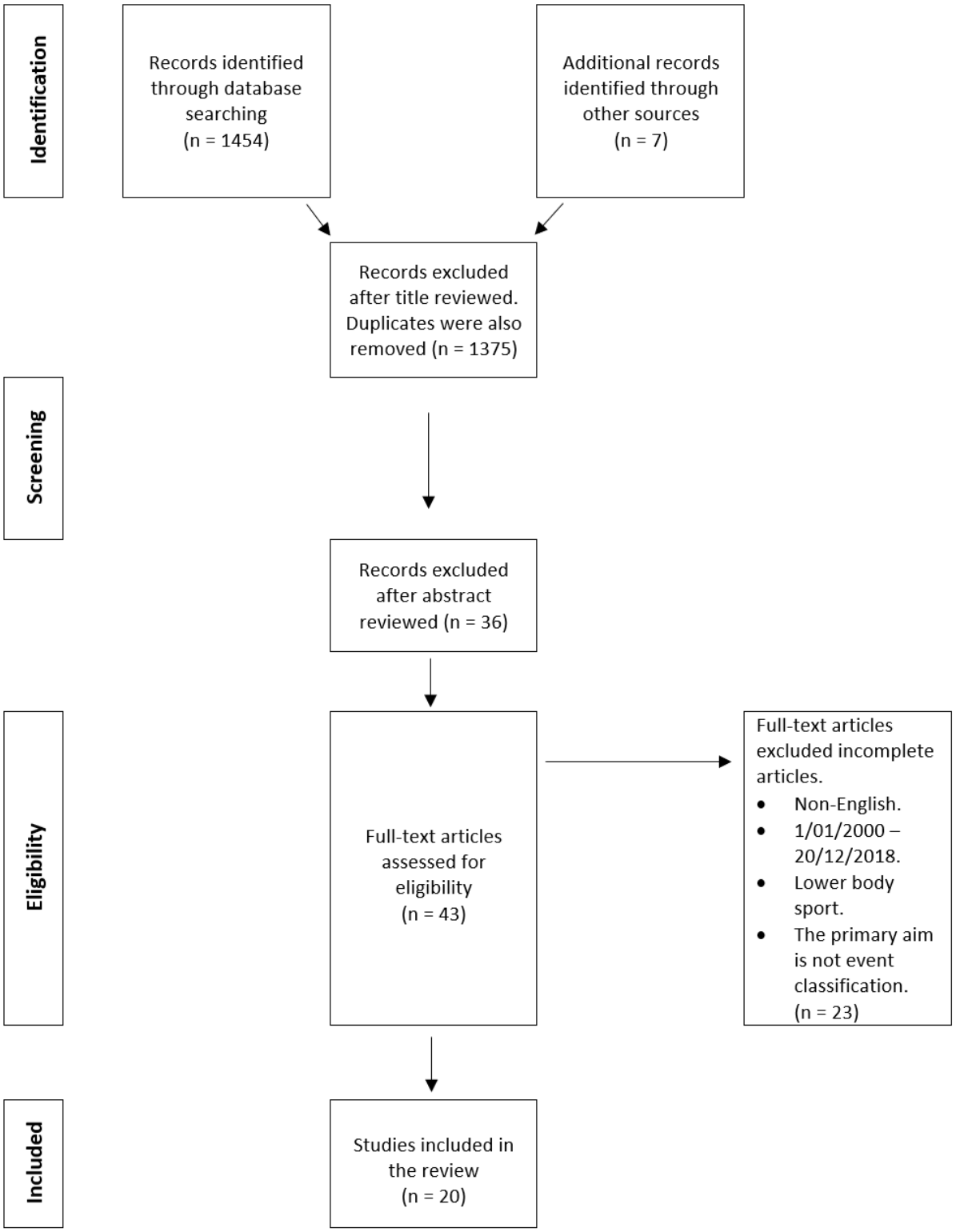


Table 2-2 summarises the study characteristics from the 20 research articles. Eight out of the 20 studies examined stroke classification in a racket sport (tennis,[48,55-60] badminton,[55,61] and squash[55]), while the remaining studies looked at classification in cricket fast bowling,[62-64] cricket batting,[65] baseball,[66,67] hockey,[51] volleyball,[67] beach volleyball,[68] and basketball.[49,52,69,70] Eight studies included both novice and experienced athletes, while the other studies used either novice participants (n= 1), experienced participants (n = 6), developmental athletes (n = 2) or did not state experience levels (n = 5). The study sample size ranged from 2 to 70, with the median being 12 participants.

Table 2-2: Summary of past studies that used IMUs to quantify sport-specific movements.

Reference	Study aim and Participants	IMU	Objectives (O)	Data signal processing (SP), Segmentation (S), Event detection (ED)	Feature computation (FC), Validation method (VM), Classification method (CM)	Results
Anand et al. (2017)[55]	Stoke detection and classification in tennis, badminton and squash. Tennis = 31 participants. Badminton = 34 participants. Squash = 5 participants.	Samsung smartwatch Gear S2. Triaxial ACC (±8 g) and GYR (±2000 deg·s^{-1}). SF = 100 Hz. IMU mounted on the dominant wrist.	O1: Stroke detection in all 3 sports. O2: Stoke classification (Tennis, 5 strokes; Badminton, 4 stokes; Squash, 2 strokes).	SP: Not specified. S: Event detection window (size not specified). ED: A combination of ACC and GYR thresholds, jerk based detection and shape-based detection of the time warp signal.	FC: ~2000 time domain and correlation-based. Reduced to an unspecified number by feature selection. VM: Two independent groups. CM: O1: Refer to ED; O2: Machine learning (LR, DCNN, BLSTM).	O1: F-score: Tennis = 92%, Badminton = 88%, Squash = 96%. O2: Average accuracy across sports: LR = 86.1%, DCNN = 88.1%, BLSTM = 88.9%.
Bai et al. (2016)[52]	Classification of shooting attempts in a one-to-one basketball game. 2 participants.	Microsoft Band & Android phone. Triaxial ACC, GYR. SF = 32 Hz & 100 Hz respectively. Microsoft Band device mounted on the dominant wrist and an Android phone placed in the trouser pocket.	O1: Detecting shot event. O2: Detecting shooter.	SP: Not Specified. S: Sliding window (length individualised to the player, 4-6 s, 1 s overlap). ED: N/A.	FC: Time and frequency domain features for Microsoft Band (66) and Android phone (15). VM: 10-fold cross-validation. CM: O1: Machine learning (RF, NB, CT, SVM, KNN); O2: Machine learning (RF, SVM, KNN).	O1: RF had the highest accuracy (88%). O2: SVM had the highest F-score (92.80%).
Connaghan et al. (2011)[48]	Stroke classification in tennis. 8 participants of different playing abilities.	TennisSense IMU system. Triaxial ACC ± 10 g, GYR ± 1200 deg·s^{-1} and MAG ± 6 Gauss. SF = not specified. IMU mounted distally on the dorsal surface of the forearm.	O1: Stroke and non-stroke detection during a competitive match. O2: Determining the best sensor modality for classifying data into 3 main strokes.	SP: Data was filtered for noise (filter not stated). S: Event detection window (size not specified). ED: ACC magnitude > 3 g.	FC: Unspecified. VM: O1: 10-fold cross-validation; O2: LOOCV. CM: Machine learning (NB).	O1: Accuracy = 85%, F-score = 77.61%. O2: Individually, the ACC had the highest accuracy (79%). A combination of all three sensors gave the best accuracy (90%).
Hölzemann and Van Laerhoven (2018)[69]	Classification of activities in basketball. 3 experienced participants (M).	Custom IMU (MPU-9250). Triaxial ACC. SF = 200 Hz (down-sampled to 25 Hz). IMU mounted on the wrist.	O1: Classify 4 movement activities related to basketball.	SP: Not specified. S: Sliding window (1 s). ED: N/A.	FC: 6 time-domain features. VM: LOOCV. CM: Machine learning (KNN, RF).	O1: RF had the highest accuracy (87.48%) and F-score (69.45%).

Kautz et al. (2017)[68]	Stroke classification in beach volleyball. 30 participants of different playing abilities. (M&F).	Bosch BMA280 IMU. Triaxial ACC (±16 g). SF = 39 Hz. GYR was not used. IMU mounted on the dorsal surface of the wrist of the dominant hand.	O1: Classifying data into 9 beach volleyball actions (1 null class). O2: Determine if accuracy improves with feature selection.	SP: High-pass Butterworth filter (8 Hz), then a low-pass Butterworth filter (3 Hz). S: Event detection window (4 s). ED: Max ACC signal > 5.1 $m \cdot s^{-2}$ & absolute ACC 200 ms before maximum.	FC: O1: 39 time and frequency domain features; O2: Reduced to 12 selected features by feature selection. VM: 10-fold cross-validation. CM: Machine learning (SVM, KNN, NB, CT, RF, VOTE, DCNN).	O1: DCNN had highest balanced accuracy (79.5%). SVM = 59.7%. O2: Accuracy with feature selection was consistently lower.
Khan et al. (2017)[65]	Classification of different cricket batting shots. 6 participants of different playing abilities (M&F).	4 IMUs. Triaxial ACC, GYR, MAG. SF = 100 Hz. IMUs mounted on the lower part of all 4 limbs.	O1: Five different levels of classification. 1 = Shot or no shot, 2 = hit of miss, 3 = shot direction, 4 = type of footwork used, and 5 = shot type.	SP: Not specified. S: Sliding window (1.6 s). ED: N/A.	FC: 464 time and frequency domain features. VM: LOOCV. CM: Machine learning (CT, KNN, SVM).	O1: No significant difference between type of classifier used and level of classification. Average F-score across all levels: SVM = 84.14%, KNN = 83.26%, DT = 79.24%.
Kos and Kramberger (2017)[56]	Stroke classification in tennis. 7 participants of different playing abilities.	IMU: Triaxial ACC (±16 g) and GYR (±2000 $deg \cdot s^{-1}$). SF = 1000 Hz. IMU was mounted on the posterior surface of the wrist.	O1: Classify into 3 main strokes.	SP: Not specified. S: Event detection window (size not specified). ED: Derivative average of all 3 ACC axes > 300.	FC: 4 time-domain features. VM: Not specified. CM: Manually defined algorithm (GYR max in y axis = backhand; GYR max in x & min in z = serve; GYR max in x & min in y + min in z < 1500 $deg \cdot s^{-1}$= serve, > 1500 $deg \cdot s^{-1}$= forehand).	O1: Accuracy: Serve = 98.8%, Forehand = 93.5% and backhand = 98.6%.
Kos et al. (2016)[57]	Stroke classification in tennis. 3 players of different playing abilities.	IMU: Triaxial ACC (±16 g) and GYR (±2000 $deg \cdot s^{-1}$). SF = 1000 Hz. IMU was mounted on the posterior surface of the wrist.	O1: Classify into 3 main strokes.	SP: Not specified. S: Event detection window (size not specified). ED: Derivative average of all three GYR axes > determined threshold.	FC: 5 time-domain features. VM: Not specified. CM: Manually defined algorithm (GYR max in y axis & min in x = backhand; GYR max in x & min in z = serve; GYR max in x & min in y = forehand).	O1: Accuracy: Serve = 100%, Forehand = 96.23% and backhand = 98.11%.
Mangiarotti et al. (2019)[70]	Classification of activities in basketball between two participants.	2 Custom IMU (MPU-6050). Triaxial ACC ± 16 g and GYR ±2000 $deg \cdot s^{-1}$. SF = 250 Hz.	O1: Classify 3 movement activities related to basketball.	SP: Butterworth low pass filter. S: Sliding window (1 s, 80% overlap).	FC: Unspecified number of time-domain features. VM: Non-independent training and testing group.	O1: F-score = 94.56%.

		IMUs were mounted on the posterior surface of the wrist.		ED: N/A.	CM: Machine learning (Combination of KNN and SVM).	
McGrath et al. (2019)[63]	Fast bowling detection in cricket. 17 experienced participants (M).	Sabel Sense IMU. Triaxial ACC ± 16 g and GYR ±2000 deg·s^{-1}. SF = 250 Hz. IMU was mounted to the upper posterior trunk between the shoulder blades.	O1: Bowling detection. O2: Determine the best window size. O3: Bowling detection with sampling frequency down-sampled to 125 Hz, 50 Hz and 25 Hz.	SP: 1Hz Butterworth low pass filter. S: Event detection window (10 s and 1 s). ED: Peaks > 500 deg·s^{-1} in the magnitude of GYR x, y and z axis.	FC: 282 time and frequency domain features. Reduced to 223 by feature selection. VM: Two groups (6 new participants in the testing group, 4 the same) CM: Machine learning (RF, SVM, PSVM, NN, XGB).	O1: F-score: SVM & PSVM = 100%. O2: No significant difference between 10 s and 1 s for any classifiers. O3: F-score: No significant difference at 25 Hz. XGB and RF = 97%.
McNamara et al. (2015)[62]	Fast bowling detection in cricket. Training phase: 12 experienced fast bowlers (M). Competition phase: 5 experienced fast bowlers (M).	Catapult MinimaxX S4 IMU. Triaxial ACC ± 10 g, GYR ± 1200 deg·s^{-}, and MAG. SF = 100 Hz. IMU was mounted to the upper posterior trunk between the shoulder blades.	O1: Classifying training data into bowling and non-bowling events. O2: Classifying competition data into bowling and non-bowling events.	SP & S not stated. ED: See CM.	FC: Not specified. VM: Not specified. CM: Manually defined algorithm. Back-foot contact (ACC) and peaks in the rotation speed (GYR) of the upper torso (> 500 deg·s^{-1}).	O1: Sensitivity = 99%, specificity = 98.1%. O2: Sensitivity = 99.5%, specificity = 74%.
Mitchell et al. (2013)[51]	Classification of activities in five-a-side soccer and field hockey using an IMU from a smartphone. Soccer = 15 amateur participants. Hockey = 17 elite participants.	1 Google Nexus One or HTC Desire smartphone. Triaxial ACC. SF = 16 Hz & 25 Hz respectively. IMU was mounted to the upper posterior trunk between the shoulder blades.	O1: Classifying data into 7 activities common to both sports. O2: Determining the best window length for both sports. O3: Determine the best mother wavelet function and DWT level.	SP: Filter designed by DWT. S: Event detection window (varying window length, 1 to 9 s). ED: Not specified.	FC: 42 descriptive features extracted utilising DWT. VM: Repeated (10x) cross-validation (66/33% train/test split). CM: Machine learning (SVM, KNN, NB, CT, NN).	O1: Hockey: NN had the highest F-score (82.3%), Soccer: NB (79.9%). Combination: Hockey = 88.8%, Soccer = 86.3%. O2: Hockey 7 s, Soccer 3 s. O3: Wavelet. Hockey = bior1.1, Soccer = rbio1.1 (Level 6 for both).
Mlakar and Luštrek (2017)[58]	Stroke classification in tennis. 5 experienced participants.	Catapult S5. GPS = 10 Hz, triaxial ACC ± 16 g, GYR ± 2000 deg·s^{-}, and MAG. SF = 100 Hz. IMU was mounted to the upper posterior trunk between the shoulder blades.	O1: Shot detection. O2: Classify into 3 main strokes.	SP: Butterworth bandpass filter (1.5 & 25 Hz) applied to windowed data. S: Sliding window (0.8 s and 1.2 s). ED: N/A.	FC: Unspecified number of time-based features. VM: LOOCV. CM: O1 & O2: Machine learning (RF).	O1: F-score = 94.1% O2: F-score: Forehand = 91%, Backhand = 92.1%, Serve = 99%.

Murray et al. (2016)[66]	Pitching and throwing detection in baseball. 17 development athletes.	Catapult MinimaxX S4 IMU. Triaxial ACC ± 10 g, GYR ± 1200 deg·s⁻, and MAG. SF = 100 Hz. IMU was mounted to the upper posterior trunk between the shoulder blades.	O1: Classifying training data into bowling and non-bowling events. O2: Classifying competition data into bowling and non-bowling events.	SP & S not stated. ED: See CM.	FC: Not specified. VM: Not specified. CM: Manually defined algorithm. Unspecified peaks in the rotation velocity (GYR), unspecified ACC measures.	O1: Sensitivity = 100%, specificity = 79.8%. O2: Sensitivity = 100%, specificity = 74.4%.
Nguyen et al. (2015)[49]	Classification of activities in basketball. 3 participants.	5 custom IMUs (LSM9DS0). Triaxial ACC. SF = 200 Hz (down-sampled to 40 Hz). IMUs mounted on the lower back, both feet and both knees.	O1: Classify 7 movement activities related to basketball.	SP: Low pass filter (15 Hz). S: Sliding window (3.2 s, 50% overlap). ED: ACC Z axis > 4 g.	FC: 80 time and frequency domain features. An unspecified number of cross-correlation coefficients. VM: LOOCV. CM: Machine learning (SVM).	O1: Overall accuracy = ~67.2%.
Rawashdeh et al. (2016)[67]	Classification of volleyball serves and baseball throws. 11 participants of different playing abilities.	Custom made IMU. Triaxial ACC ± 16 g, GYR ± 2000 deg·s^{-1}, MAG ± 8 Gauss. SF = 50 Hz. IMU was mounted to the medial surface of the upper arm of the dominant arm.	O1: Classifying data into volleyball serves and baseball throws. Non-events included shoulder rehab exercises.	SP: No specified. S: Event detection window (size not specified). ED: Event detection: Elevation of Arm > 45 deg and Angular Rate > 400 deg·s^{-1}.	FC: 81 time and frequency domain features. Reduced to 8 features by feature selection. VM: Cross-validation (40/60% train/test split). CM: Machine learning (DT).	O1: Accuracy = 86%.
Salman et al. (2017)[64]	Legality analysis of fast bowling in cricket. 14 medium to fast-paced bowlers (M).	3 Custom IMU. Triaxial ACC and GYR. SF = 150 Hz. IMU was mounted on the upper arm, forearm, and wrist.	O1: Classify the legality of bowling action. O2: Determine if accuracy improves with feature selection.	SP: Filter not stated. S: Event detection window (7 s). ED: GYR maxima in the x-axis.	FC: 120 time-domain features. Reduced to 21 by feature selection. VM: LOOCV. CM: Machine learning (SVM, K-NN, NB, RF, NN).	O1: F-score: NB = 80.7%, SVM = 89%, KNN = 90.3%, NN = 87%, RF = 86%. O2: Highest F-score was obtained after feature selection for all classifiers.
Srivastava et al. (2015)[59]	Stroke classification in tennis. 14 participants of different playing abilities.	Samsung smartwatch Gear S. Triaxial ACC (±8 g) and GYR (±2000 deg·s^{-1}). SF = 25 Hz. IMU mounted on the wrist.	O1: Classify into 3 main strokes. O2: Classify 3 main sub-strokes.	SP: Pan Tomkin's algorithm was used to isolate shot signal from noise. S: Sliding window (3*sampling rate). ED: ACC spike above an unspecified threshold.	FC: Not specified. VM: Not specified. CM: Machine learning (Phase 1: DTW. Phase 2: QDTW).	O1: Sensitivity: Professional = 99.6%, Novice = 99.3%. O2: Sensitivity: Professional = 90.7%, Novice = 86.2%.

Wang et al. (2016)[61]	Stroke classification in badminton. 12 club participants.	4 IMUs. Triaxial ACC ± 16 g, GYR ± 500 deg·s^{-1}, biaxial GYR ± 500 deg·s^{-1}. SF = 500 Hz. IMUs mounted to both wrists, waist, and right ankle.	O1: Classifying data into the 14 strokes and 4 non-stoke motions during a competitive match. O2: Determining the best segmentation approach.	SP: Low pass filter designed from a Wavelet function Coif5. S: Sliding window (1 s, 50% overlap) vs sliding window with high acceleration (> 1.5 g) vs event detection window (1 s). ED: ACC magnitude > 1.5 g & > 1 s removed from adjacent max.	FC: 120 time-domain and frequency-domain features. VM: 4-fold cross-validation. CM: Machine learning (HMM, NB, CT, SVM, LDF).	O1: HMM had the highest accuracy (97.96%). O2: The event detection window had the highest accuracy (97.96%).
Whiteside et al. (2017)[60]	Stroke classification in tennis. 19 development athletes (M&F).	IMeasureU IMU. Triaxial ACC ± 16 g, GYR ± 2000 deg·s^{-1}. SF = 500 Hz. IMU was mounted distally on the ventral surface of the forearm of the hitting limb.	O1: Classifying data into the 3 main stokes and false stroke. O2: Classifying data into 8 strokes and false stroke.	SP: Linear interpolation (smoothed using a 50-point moving average). S: Event detection window (1 s). ED: Max ACC signal > 25 m·s^{-2} & > 1.25 s removed from adjacent max. GYR > 400 deg·s^{-1} within 0.06 s either side of ACC max.	FC: 40 features (5 features x 8 waveforms). VM: 10-fold cross-validation. CM: Machine learning (SVM, DA, RF, KNN, CT, NN).	O1 & 2: SVM (cubic kernel) had the highest overall F-score (O1 = 94.33%, O2 = 90.01%).

Key: ACC = Accelerometer; bior1.1 = Biorthogonal 1.1; BLSTM = Bidirectional long short-term memory networks; DCNN = Deep convolutional neural networks; CT = Classification tree; DA = Discriminant analysis; DTW = Dynamic time warping; DWT = Discrete wavelet transform; F = Female; FBC = Feature based classification; GYR = Gyroscope; HMM = Hidden Markov model; KNN = K-nearest neighbour; LDF = Linear discriminant function; LOOCV = Leave one out cross-validation; LR = Logistic regression; MAG = Magnetometer; M = Male; N/A = Non-applicable; NB = Naive Bayesian classifiers; NN = Neural network; PSVM = Polynomial SVM; QDTW = Quaternion dynamic time warping; a_{rad} = Radial acceleration; RBF = Radial basis function; RF = Random forest; rbio 1.1 = Reverse biorthogonal 1.1 wavelet; SF = Sample frequency; SVM = Support vector machine; ω_{tan} = Tangential angular velocity; VOTE = Plurality voting; VM = Validation method; XGB = Gradient boosting.

Discussion

Evaluation metrics

Included studies used a range of metrics to evaluate the performance of their classification models. The classification of a binary event has four possible outcomes: (1) true positive (TP), where the event is labelled correctly as an event by the algorithm; (2) true negative (TN), where a non-event is labelled correctly as a non-event by the algorithm; (3) false positive (FP), where a non-event is incorrectly labelled as an event by the algorithm; and (4) false negative (FN), where an event is incorrectly labelled as a non-event by the algorithm. These outcomes can be presented in several ways, but commonly used methods include sensitivity (also known as recall), specificity (also known as the true negative rate), precision (also known as the positive predictive value), accuracy (the fraction of correctly identified events and non-events), F-score (the harmonic mean of precision and sensitivity), and balanced accuracy (the mean of sensitivity and specificity). Table 2-3 provides formulas and a description of each metric. Thirteen studies reported (or displayed classification matrices so the reader could calculate) more than one evaluation metric.[51,52,58,60,62-65,67-70] Depending on the nature of the classification problem, reporting only accuracy can misrepresent the results; data with imbalanced classes (e.g. a low number of observations in one activity class relative to another) can lead to inflated overall accuracy. Balanced accuracy and F-score are preferred in this regard. While the acceptable level of accuracy may depend on the application (e.g., a higher error rate might be acceptable for a system that predicts step counts for general consumers but unacceptable when quantifying bowling frequency in elite cricket players), the goal of fitting classification models is to achieve the best accuracy possible. An accurate classification model is particularly important when there is a relationship between activity parameters (e.g., load, volume, technique) and injury or performance.

Table 2-3: Description of the performance measures used.

Format	Formula	Description
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	The fraction of correctly identified events and non-events.
Sensitivity	$\frac{TP}{TP + FN}$	The proportion of correctly identified events, regardless of all non-events.
Specificity (Recall)	$\frac{TN}{TN + FP}$	The proportion of correctly identified non-events, regardless of all events.
Precision	$\frac{TP}{TP + FP}$	The proportion of correctly identified events relative to the total number of events detected.
F-score	$\frac{2 \text{ x precision x sensitivity}}{\text{precision} + \text{sensitivity}}$	The harmonic mean of precision and sensitivity.
Balanced accuracy	$\frac{\text{sensitivity} + \text{specificity}}{2}$	The mean of sensitivity and specificity.

Key: FN = False negative; FP = False positive; TN = True negative; TP = True positive.

IMU number

Using multiple IMUs means that sensors can be attached to both the trunk and extremities. This may provide complementary information to classification algorithms, therefore improving accuracy.[45] In this review, six studies[49,52,61,64,65,70] used multiple IMUs (between 2 and 5). Only one study examined individual versus combined accuracy and found that accuracy from a single IMU on the playing wrist was high (95%) when classifying four main badminton shots (forehand serve, backhand serve, forehand clear, and backhand clear).[61] However, when classifying ten additional sub-shots, the accuracy was significantly reduced (76%). When using a combination of four IMUs to classify the main and sub-shots, improved accuracy of 97% was observed. This may have been one of the reasons for the reduced accuracy when using a single IMU for classifying sub-shots in tennis,[59,60] compared to the results obtained by classifying main shots. Although multiple IMUs may improve classification performance, there are limitations regarding cost, simplicity of use, computational demand, movement restriction, and comfort, meaning a single IMU may still be preferred in a given context.

IMU specification

The three main specifications commonly reported for an IMU include: sampling frequency; the accelerometer, gyroscope, and magnetometer measurement range; and the number of axes each sensor can record. Triaxial IMUs can capture more data, improving classification accuracy, particularly for limb-based movements.[71] Sampling frequency ranged from 16 Hz to 1000 Hz in the reviewed studies, and sensor measurement range varied from 8 to 16 g for the accelerometer and 500 to 2000 degrees per second for the gyroscope. Although studies have shown that accuracy can be high when using a low sampling rate IMU,[51,59,63,68] only one study compared the accuracy of a high versus low sampling rate when detecting fast bowling events in cricket.[63] The accuracy did not significantly decrease when data were down-sampled from 250 Hz to 25 Hz (F-score 1.0 vs 0.97). It is important to note that this classification problem only had two possible outcomes (bowl or non-bowl), and the IMU was attached to the torso. Attaching an IMU with a low sampling rate or measurement range to an extremity may mean that a significant amount of information is lost during high-velocity movements. For example, a cricket bowl can generate upwards of 80 g at the wrist, but many IMUs are limited to 16 g. It is currently unclear how this impacts classification accuracy for different movements. Future studies should compare the minimum specifications required to inform the appropriate choice of IMU for each movement.

Lastly, it should also be mentioned that sensor resolution, which is seldom reported, can play an essential role in the accuracy of recorded data. IMU resolution refers to the smallest change that the sensor can detect in the activity that it is measuring. It is influenced by the full-scale range of the IMU.

For example, a higher resolution is desired when slight changes in acceleration are of interest without limiting the full range of the accelerometer. Most standard IMU units will have a minimum 8-bit resolution, with the vast majority being 16-bit or above.

IMU location

IMUs can be positioned at several locations on the body. This includes anywhere on the torso and all four extremities. The optimal placement location is likely specific to the sport under study.[72] Five studies mounted an IMU solely on the participants' torso,[51,58,62,63,66] while the remaining 15 studies had at least one IMU located on an extremity (e.g. wrist or ankle). A common location for the torso is between the shoulder blades on the upper thoracic spine.[51,58,62,63,66] The benefit of this location is that it does not impede movement, is protected from more common frontal impacts, and can be embedded in clothing.[60,73] These reasons make it a popular choice for sports that involve high impact forces on the body (e.g. rugby). It is also a convenient location for IMUs with lower accelerometer and gyroscope maximum ranges, as there is generally less acceleration and angular velocity than in a limb-based location.[51,63] Alternatively, IMUs with higher dynamic ranges have been successfully attached to extremities in sports with significant limb movement, such as tennis[60] and badminton.[61] These locations may provide an improved classification of limb movements, as the trunk is predominately exposed to linear acceleration and may not provide adequate spatiotemporal information to distinguish limb-based events.[74] This was evident in a study that examined badminton stroke classification,[61] which showed a wrist-mounted IMU of the dominant hand had an overall accuracy of 81.6% compared to 42.6% when placed on the waist. As IMUs with higher sampling frequency and measurement ranges are embedded in affordable wrist-worn consumer-grade products (e.g., the Apple Watch), developing classification algorithms for these systems may increase the accessibility to accurate activity classification.

Sensor modality

Activity classification algorithms can use data from just one sensor (i.e., the accelerometer, gyroscope, or magnetometer) or several sensors simultaneously. The magnetometer is used the least frequently of the three sensors present in most IMUs. This is because a controlled magnetic field without any distortion is required for optimal results,[75] especially with fast linear motions.[74] Two studies,[48,65] used data from all three sensors (accelerometer, gyroscope and magnetometer), while the remaining studies used either a combination of the accelerometer or gyroscope[52,56-64,66,67,70,76] or only the accelerometer[49,51,68,69]. Only one study investigated different combinations of sensors for classifying tennis strokes.[48] The accelerometer showed slightly higher classification accuracy (79%) compared to the gyroscope (76%) and the magnetometer data (76%), but 90% accuracy was achieved when combining all three sensors. The authors proposed that the lower accuracy observed using gyroscope data could be related to high variation in the temporal orientation of a given tennis stroke due to

differing skill levels between participants. The importance of the gyroscope may depend on the movement being performed, the IMU placement position, and the gyroscope features that are quantified (see section 'Feature engineering and feature selection'). McGrath et al. (2019)[63] showed that when using a trunk-mounted sensor, gyroscope features consistently had higher importance for all classification models when detecting cricket bowls. This could have been because trunk rotation patterns are more distinct between activities (bowl and throw) compared to acceleration patterns. The low classification accuracy of volleyball strokes observed by Kautz et al. (2017)[68] may have been due to the omission of gyroscope data, as rotational patterns of different volleyball strokes are likely to be distinct. Although most studies do not report feature importance or compare sensor modalities, it is likely that accelerometer and gyroscope features are equally important for classifying movements in sports that have high rotational velocity.

Pre-processing

Pre-processing is an important step in the classification process, as it can improve accuracy and reduce the computational load for the classifier.[77,78] Pre-processing generally involves filtering raw data, data segmentation, feature engineering, and feature selection.

Filtering

Filtering makes raw data more representative of the desired activity by minimising the amount of data not associated with human movement tasks. This is often called signal artefact or noise and is usually caused by clothing, skin, muscle and other subcutaneous tissue movements.[79] A wide range of algorithmic approaches can convert raw data to time, frequency and discrete domains of representation so that it is easier to apply different filtering techniques.[77] Several filters have been specified, including moving average filters,[60] wavelet functions,[51,61] low pass filters,[49,58,63,68,70] and high pass filters.[58,68] Wavelet functions allow time-frequency characteristics to be examined, which help detect events involving a non-repetitive movement that varies over time (e.g. cricket batting).[78] Furthermore, wavelet functions can capture sudden changes in signals that are common in accelerometer and gyroscope readings.[77] There were no studies in this review that compared different filtering techniques. However, a low pass filter is the most common filter as the artefact is typically represented as high-frequency signals.[78] IMU filtering techniques must be not over-aggressive and preserve signal characteristics relevant to the activity of interest.[45]

Data segmentation

Data segmentation involves arranging data into windows (segments of data), so that features can be extracted. Despite the potentially large impact that different window types and sizes can have on classification accuracy,[45,71] the comparison of window techniques is rarely performed,[67,80] especially when classifying sporting events.

The sliding window technique is a common segmentation approach that requires no prior identification of potential events. Signals are divided into fixed-length windows that can be arranged sequentially or overlapping. A total of eight studies used this approach, in cricket,[65] basketball,[49,52,69,70] tennis,[58,59] and badminton.[61] Four of these studies used an overlapping sliding window (ranging from 16 to 50 % overlap).[49,52,61,70] This helps prevent one of the limitations of traditional sliding windows when activities occur at either end of the window segment.[61,81] Another potential limitation is a large amount of null data (data not of interest) retained. This can increase the error rate and the computation demand during the modelling process.[67] Sliding windows may be beneficial when classifying all activities within a given period, such as movements of interest and recovery periods (e.g., standing still).

An event-defined window is formed around a previously detected signal event. This windowing technique can decrease the processing time and improve model performance by only retaining data around potential events of interest. This type of window is often used in event classification involving sporadic movements,[80] and most of the studies (12 studies) used this approach. Event detection methods included using a predetermined threshold from either raw or derived accelerometer data,[48,49,56,59,61,68] raw or derived gyroscope data,[57,63,64,67] or a combination of the two.[55,60] However, a single threshold rule may not be enough to reliably detect events due to heightened noise around signal peaks and events with more than one peak. Five studies included additional rules as follows: two excluded peaks within a certain timeframe of an adjacent peak;[60,61] Kautz et al. (2017)[68] averaged absolute acceleration values of all three axes over a 200 ms interval before the maximum deflection occurred in a beach volleyball stroke; Rawashdeh et al. (2016)[67] only included volleyball and throwing events when the arm elevated above 45 degrees; and Anand et al. (2017)[55] used jerk and shaped based detection of the time warp signal for tennis, badminton and squash.

One study compared sliding and event-defined windows in badminton.[61] The event-defined window had the highest accuracy, lowest computation time and aided in describing the movement characteristics of badminton strokes due to the consistency of the event data in each window. Generally, improving event detection accuracy leads to more precise activity classification and reduces model complexity.[45,72]

Deciding on the appropriate window length to apply is another consideration. Only one study investigated the effects of different window durations (1 to 9 seconds) when using an event-defined window to classify activities common to both five-a-side football and field hockey.[51] Their results showed that the most optimal window length was 3 seconds for football and 7 seconds for hockey. When deciding on an appropriate window length, one must factor in the activity of interest and the type of window used. Choosing a too long window increases the chances of two or more activities occurring within one window and greatly increases classification difficulty.[51] A window that is too short

may mean that the complete task is not included in the window. Viewing raw data may help determine appropriate window lengths.

Feature engineering and feature selection

Feature engineering or feature computation involves identifying and extracting quantitative measures from the data either manually or automatically. Most supervised machine learning models do not create their features, so their performance is based on the quality of the features computed. For signal data, it is common to compute features in the time and frequency domain, such as mean, standard deviation, and signal power.[72,78,82] Patterns among these features are used to distinguish between different movements or activities. Fourteen studies specified the number of features used, ranging from 4 to ~2000 features. Eight studies used only time-domain features,[55-58,60,64,69,70] while seven used a combination of features in the time and frequency domain.[49,52,61,63,65,67,68] No studies compared whether time or frequency domain features were more important for classifying certain movements; however, a broad range of features from both domains are likely to be beneficial. There is not enough evidence to propose an optimal feature set for the classification of specific movements, as it likely depends on the type of activity performed, in combination with IMU specifications and placement location.

Feature selection refers to reducing the number of features in the dataset while still representing the original feature set. A dataset with many features (i.e., high dimensionality) requires more training data to model parameter estimates,[45] increases computation time and can cause overfitting.[72,83] Five studies stated that they performed feature selection.[55,63,64,67,68] There are numerous ways to perform feature selection; a common method is to remove highly correlated features.[63] However, as shown by Kautz et al. (2017),[68] selecting too few features (39 features down to 12) severely reduced accuracy for all tested models. This may lead to insufficient information for the classification algorithm. It should be noted that some machine learning algorithms (e.g. certain neural networks (NN) and random forests (RF)) can learn the relevance of individual features and may not require feature selection.[78]

Activity classification

The reviewed studies observed two general activity classification approaches: manually defined algorithms and supervised machine learning techniques. Manually defined algorithms involve human-set rules that are usually based on previous observations. Data are classified by meeting or not meeting these sets of rules. Four studies used this approach to classify cricket fast bowling,[62] baseball,[66] and tennis strokes[56,57] with a moderate to high accuracy. The two tennis studies achieved high accuracy (> 93%) when classifying three main tennis strokes. Their algorithm was based on maximum gyroscope values among all three axes. McNamara et al. (2015)[62] used peaks in upper body rotational speed (gyroscope > 500 °/s deg·s^{-1}), front foot contact (accelerometer value unspecified),

and GPS (value unspecified) to detect cricket bowls. Lastly, following a similar approach, Murray et al. (2016)[66] used unspecified gyroscope and accelerometer peaks to classify baseball pitches. Interestingly, specificity in both studies decreased when moving from a training environment to a game setting. This could be due to the unstructured nature of actions performed during competition – confusing the algorithm. Although manually defined algorithms may offer simplicity, interpretability, and reduced computational demand, future research needs to determine if this approach can distinguish between more complex movements (e.g., different tennis sub-shots).

Common supervised machine learning models used for activity classification include the support vector machine (SVM), linear discriminant analysis (LDA), logistic regression (LR), classification tree (CT), RF, k-nearest neighbour (KNN), naive Bayes (NB), and NNs. Supervised machine learning techniques were used in 16 of the studies to classify tennis,[48,55,58-60] badminton,[55,61] volleyball,[67] beach volleyball,[68] baseball,[67] cricket,[63-65] basketball,[49,52,69,70] hockey,[51] and squash.[55]

Two studies compared the difference between deep learning neural networks (DNNs) versus conventional machine learning approaches.[55,68] Like conventional NNs that mirror some of the basic principles of a biological brain, DDNs use more than one hidden layer (typically five to more than a hundred). Different DDNs have been used for activity classification, for example, long short-term memory networks (LSTM) and convolutional neural networks (CNNs). Recently, Kautz et al. (2017)[68] found significant improvement in balanced accuracy (+19.2%) when using CNNs compared to conventional machine learning approaches. In addition, Anand et al. (2017)[55] found improvements in CNNs and LSTM versus LR (88.1% & 88.9% vs 86.1%, respectively). DDNs can learn high-level features with more complexity and abstraction than NNs when using many samples to train.[84] Further work is needed to investigate the effectiveness of deep learning methods, especially with higher sampling rate IMUs, which produce more voluminous data.

By examining classification results among the reviewed studies, no single model is best for activity classification in sport. For example, the support vector machine (SVM) was effective at classifying shooting attempts in basketball,[52] cricket bowling,[63,64] and strokes in tennis,[60] badminton,[61] and cricket[65] (all > 84 % accuracy). However, it was poor at classifying movements in beach volleyball (59.7 % accuracy),[68] where movement patterns are similar to tennis and cricket (spike vs serve and bowl). This is consistent with what has been termed the 'no free lunch theorem', where machine learning models will not perform equally well on all problems.[85] Additional reasons include changes to the number of sensors used,[45] the sensor location,[45] the sampling frequency,[71] and pre-processing techniques.[45,51,68] The lack of a clear classification model is not unique to sport, as these results are consistent in activity classification involving tasks of everyday living.[45] Combining different models may also be a viable option for improving accuracy.[51,70] Mitchell et al. (2013)[51] fused the best model for

each activity in hockey and five-a-side football and improved the F-score by 6% compared to the best individual model.

When fitting machine learning models, attention must be paid to avoid overfitting. Overfitting is the development of a model so specific to the training data that it can't be generalised to new data.[86] For most models, overfitting can be avoided by performing cross-validation. This is a process where a model is tested on an independent data set, ideally a separate sample from the training data.[68,87] In this review, only Anand et al. (2017)[55] stated using two independent groups, while McGrath et al. (2019)[63] had partially independent groups (six new participants, four from the training group). If it is not feasible to have two independent groups, other methods such as *k*-fold cross-validation can be used. In this case, the sample population is split into *k* groups (folds, e.g., 10-fold), and the model is trained on all groups except one, which is reserved for testing the model. This training and testing procedure is repeated so that each group acts as the test set. The results are then averaged. Leave-one-subject-out-cross-validation (LOOCV) is a popular form of *k*-fold where *k* is equal to the number of participants. The advantage of this method is that nearly all the data (all except one participant) is used for each training iteration, and thus produces relatively stable estimates. A total of five studies[48,52,60,61,68] performed a k-fold cross-validation with *k* ranging from 4–10, while six studies performed LOOCV.[48,49,58,64,65,69] Alternately, two studies[51,67] used a train/test split, where all data are randomly split into two groups: a training group and a testing group. The disadvantage of this method is that data from the same participant could be used to train and test the model, which may inflate accuracy estimates.

Additional considerations

It is important to note that many studies included in this review used small sample sizes (average = 12, range 3–34). It was common to have few participants performing a large number of repetitions of varying movements.[48,49,52,55-58,62,65,69,70] This increases the likelihood that the classification algorithm will not generalise well to other people or populations.[88] This is especially true when classifying sporting movements due to the variability in technique among athletes.[48,89] Training a classification algorithm on elite athletes may not generalise well to novice athletes and vice versa.[48] Therefore, caution must be used when interpreting results from studies that train their algorithm using small samples without varied skill levels. A detailed description of each movement may help to alleviate these issues, but this is not always presented. For example, Murray and Black[66] did not specify the different slot angles (i.e. over the top or side arm) that participants used when performing the throwing tasks in baseball. In addition, none of the tennis studies stipulated the type of serve used (i.e., a top spin or slice serve).

Avenues of future work

When developing a classification system, numerous factors need to be considered. These include: the number of IMUs; their specification; their placement location; and pre-processing decisions such as filtering, segmentation, feature engineering, and model selection. There are many combinations of these factors in the reviewed literature, but little formal comparison among them. This means that the optimal set of decisions to classify each movement type is unclear and will likely vary depending on the movement of interest. Given these findings, there are several recommendations for future research. Work that compares the minimum IMU specifications (e.g., sampling rate), sensor modality (e.g., accelerometer vs accelerometer + gyroscope), and IMU number and placement (e.g., single vs multiple IMUs; torso vs wrist placement) will assist researchers and practitioners with the selection of the most accurate and cost-effective device setup for their needs. The comparison of different data processing techniques is also warranted. No studies compared different filtering methods or whether time or frequency domain features were more important for classifying certain movements. A formal comparison of these decisions and presenting these results (as opposed to just presenting the most accurate model) will aid the progression of this field. Comparing deep learning frameworks to more traditional supervised learning methods is also warranted, given the promising results seen in other areas. Interestingly, as the success of a classification system in sport relies on the accuracy of classifying the individual, it is surprising that no study has looked at the effectiveness of individually-trained models (i.e., trained using one subject's data) relative to the generalisability of a single model trained on many subjects' data. This could be an effective approach in classifying upper body sporting movements, particularly those where technique can differ considerably between participants.

Conclusion

This systematic literature review examined upper body activity classification using an IMU in court or field-based sports. Both manually defined rule-based algorithms and machine learning approaches have been used to classify movements across several sports, with varying degrees of success. In addition to the choice of algorithm or model, a classification system includes the number of IMUs, their specifications and placement location, and numerous data processing decisions (e.g., data filtering, data segmentation, feature engineering and selection). Due to the limited number of studies that compare these decisions, no definitive approaches for classifying upper body movements can be recommended. However, this also presents an avenue for future work; formally comparing these decisions within the classification workflow will assist in identifying the optimal set of decisions to classify each movement of interest.